Imperial College London

17 October 2024

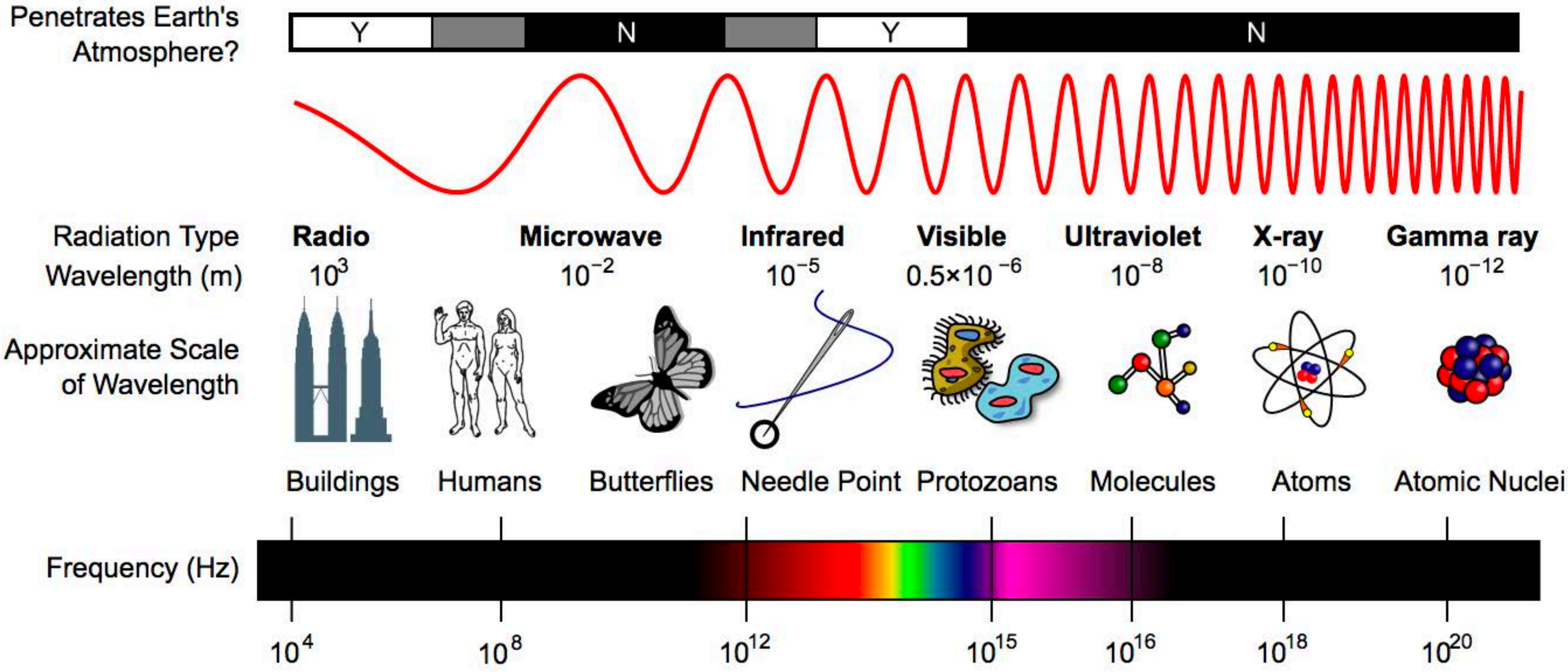
Advanced Imaging Technologies in Systems Biology and Biomedical Research

Periklis (Laki) Pantazis

What we covered

- What is light and colour?
- How does a light microscope work?

Properties of light: electromagnetic spectrum

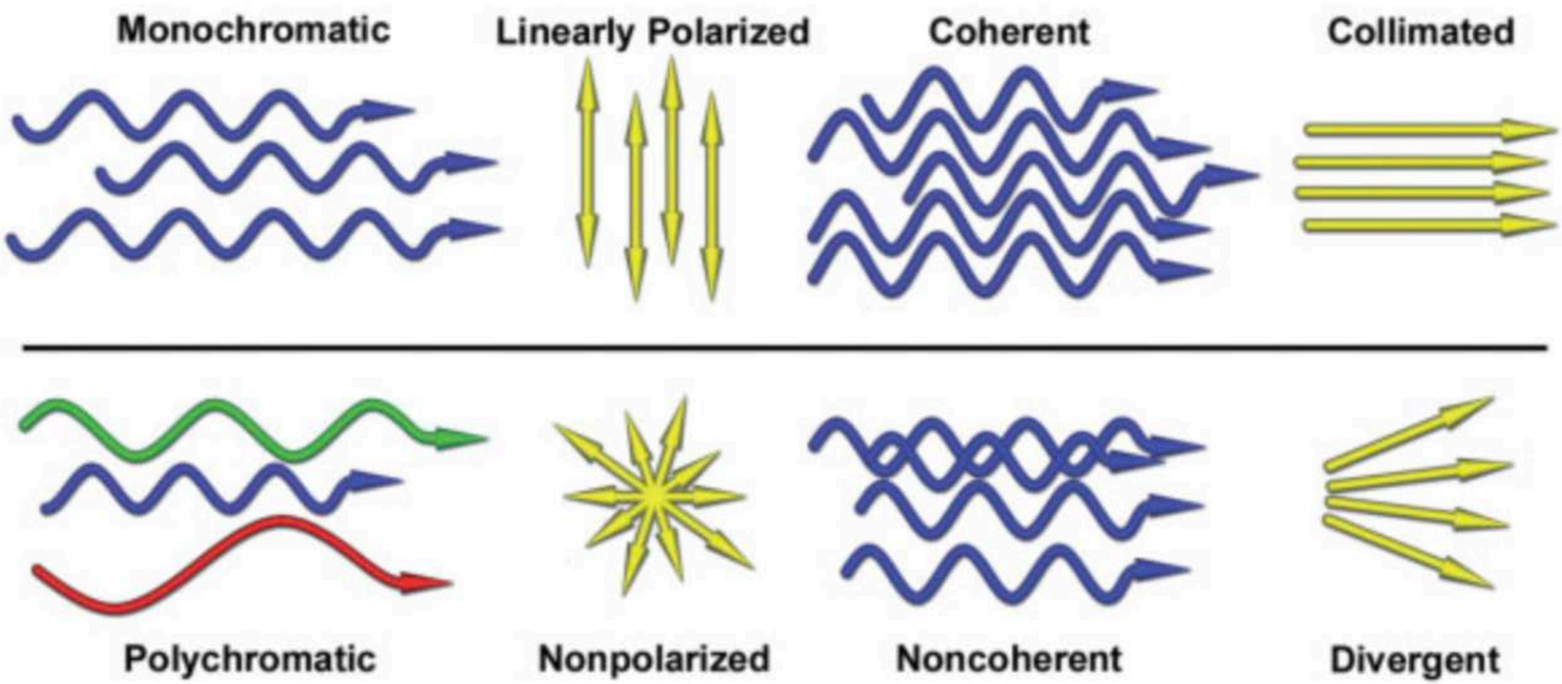


Mathematical relationship of light as particle and wave

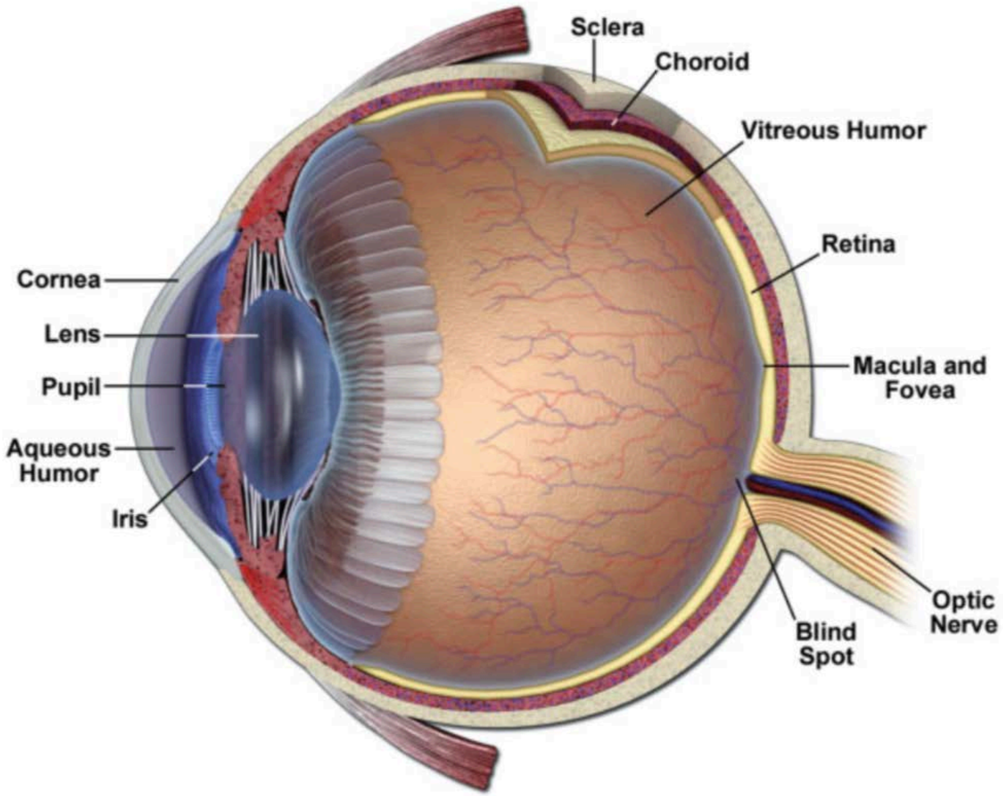
$c = v\lambda$	- denotes velocity of light
$E = hv$	- relates frequency and energy
$E = hc / \lambda$	- relates energy of a photon to its wavelength ($E = 1/\lambda$)

h = Plank's constant (6.62 x10-27erg-seconds)
c = speed of light (3x1010cm/s)
v = frequency (cycles/s)
λ = wavelength (cm)
E = energy (ergs)

The quality of light



The structure of the human eye



- Brightness

amplitude (A) of E vector

- Intensity (I)

rate of flow of light energy per unit area and per unit time across a detector surface.

- Relation of Amplitude (Energy) to Intensity (Energy Flux):

$I \sim A^2$

- Contrast

$C \sim \Delta I / I_b$ $(C=0 \text{ if } I_{obj}=I_b)$

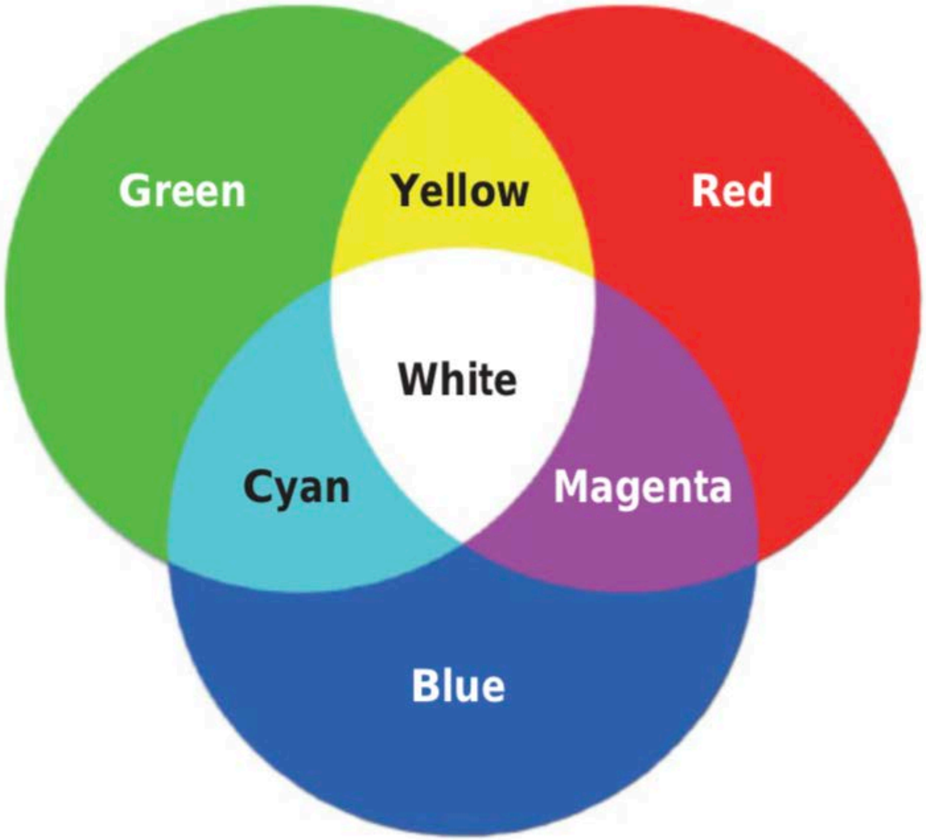
- Contrast Threshold

bright light: as little as 2-5%
dim light: 200-300%

- Eye is a Logarithmic detector

$C \sim \log_{10} (I_1/I_2)$

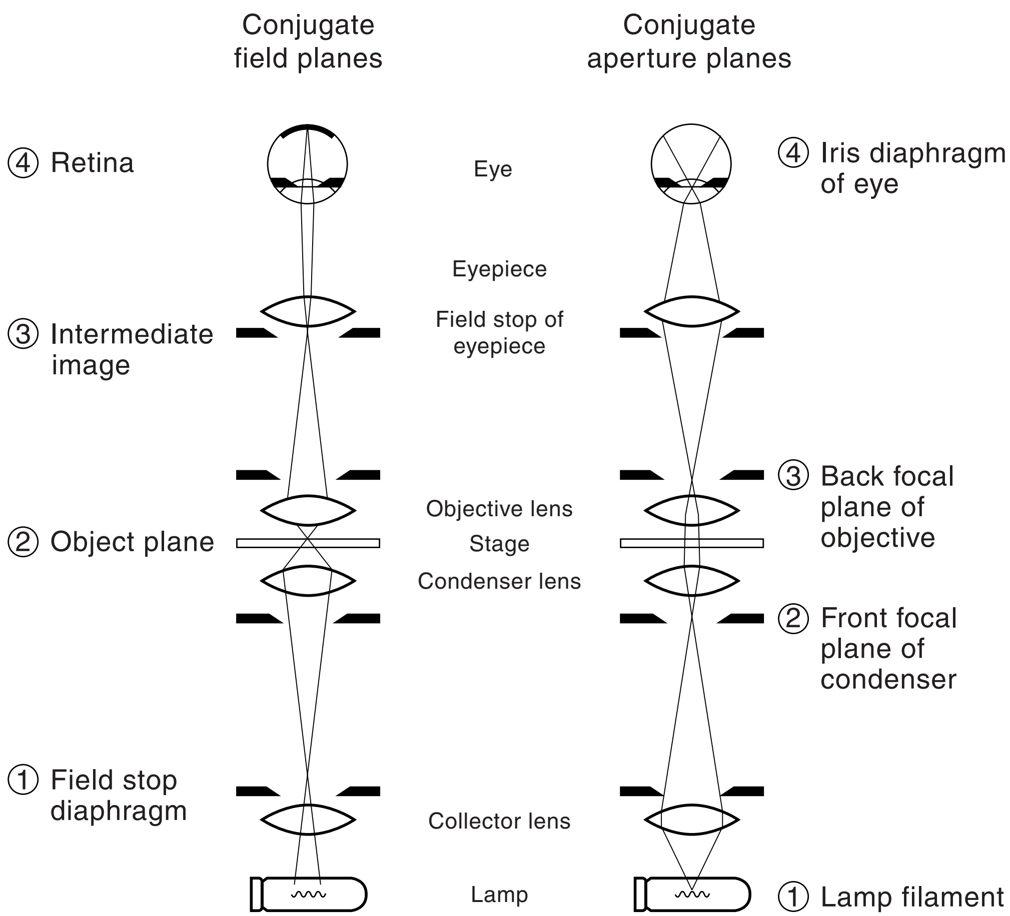
Colour vision: tristimulus system



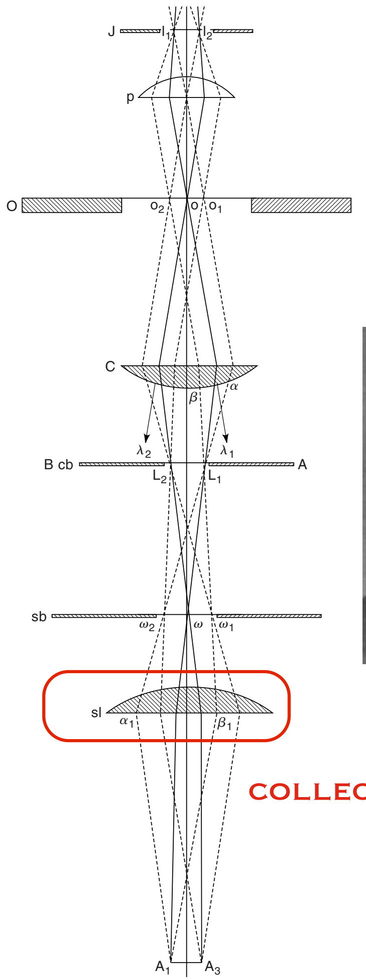
The compound light microscope

$M_{final} = M_{obj} \times M_{oc}$

Field planes and aperture planes



Köhler illumination

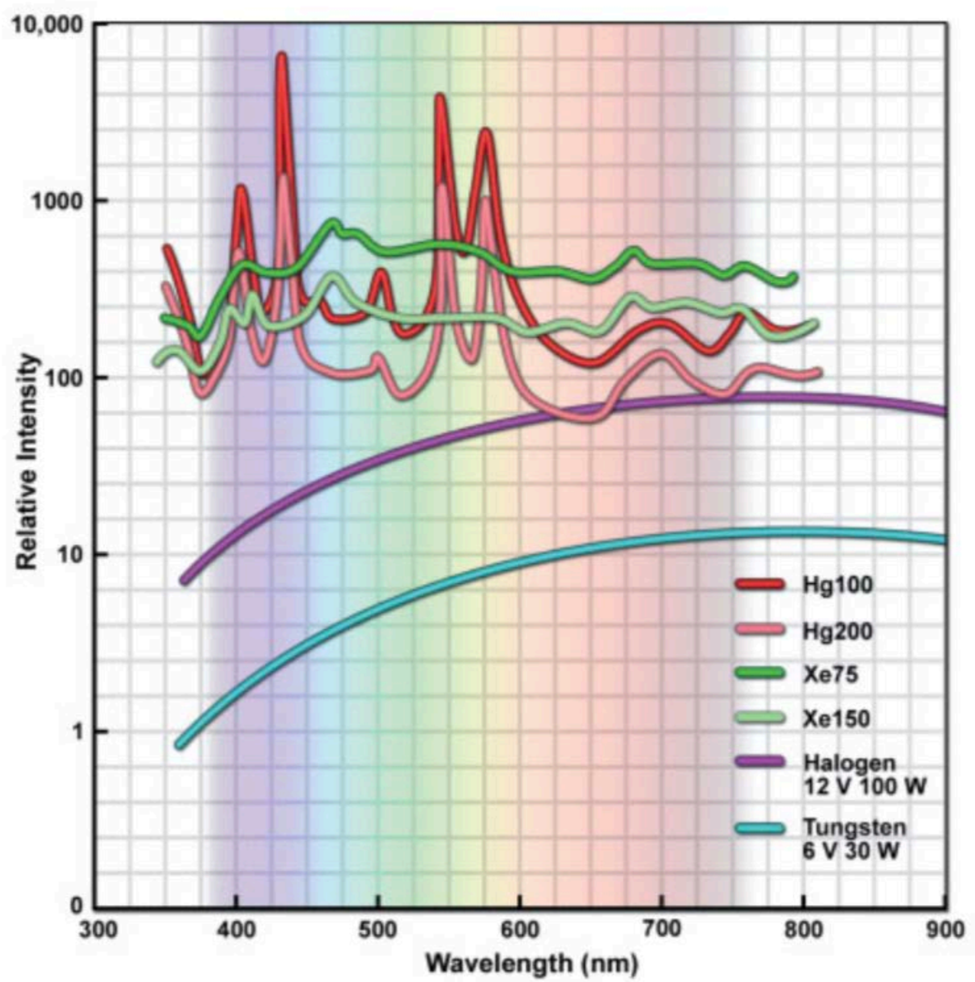


COLLECTOR LENS

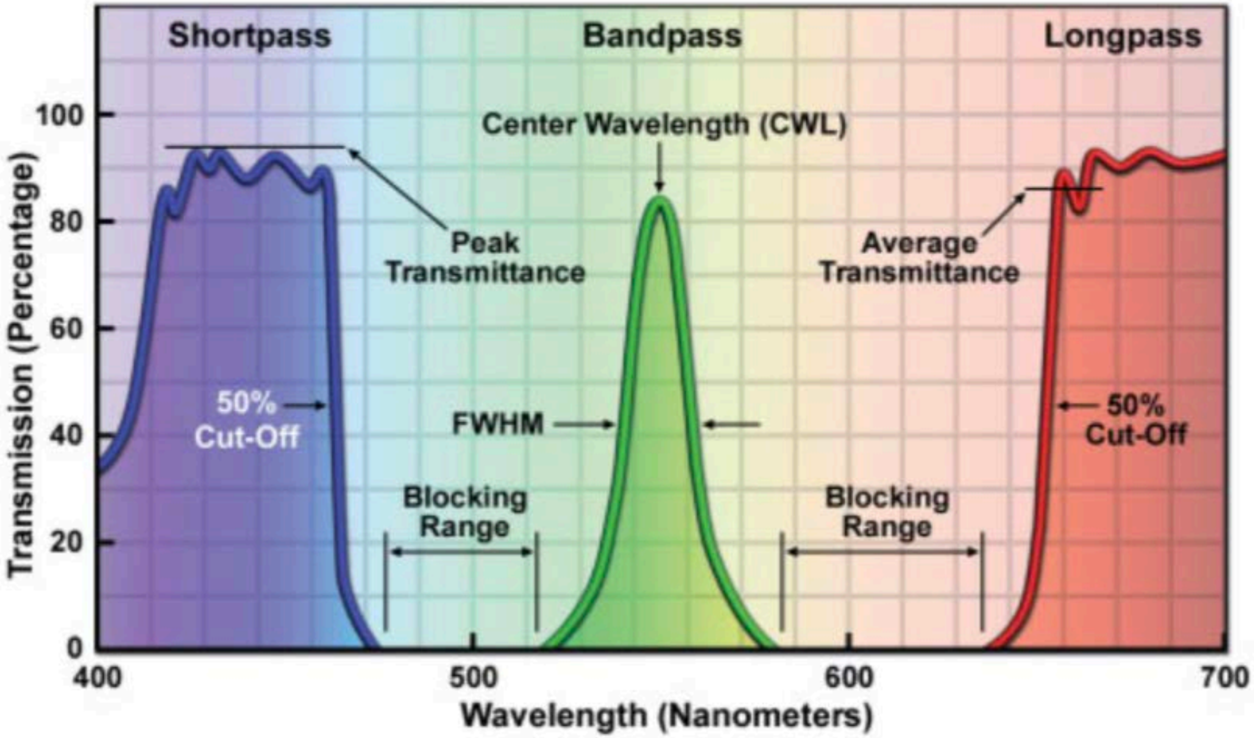
August Kohler
1866-1948

bright and even illumination

Illuminators

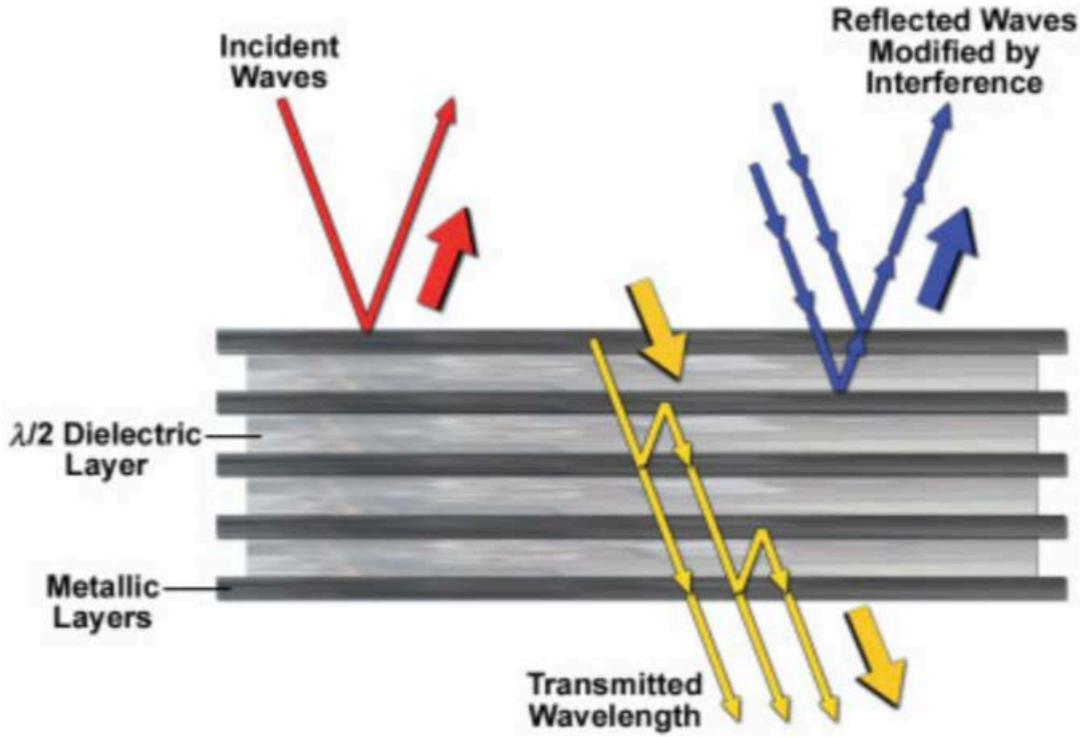


Filters for isolating the wavelength of illumination

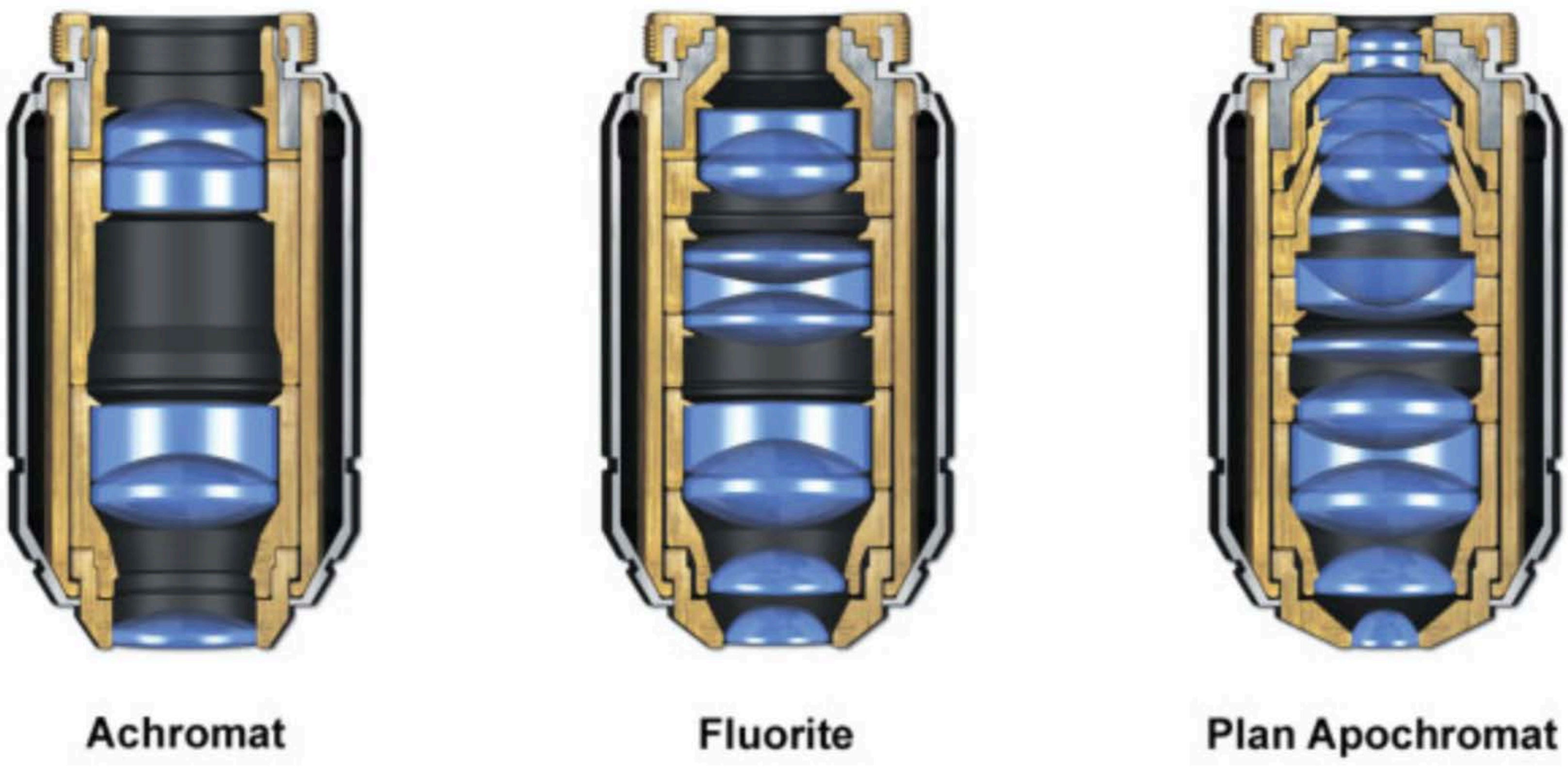


Interference filter selectively transmit waves

Interference filters



Objective designs



What we will cover

- What is fluorescence?
- How does a fluorescent microscope work?
- What are genetically encoded fluorescent proteins?

What we will cover

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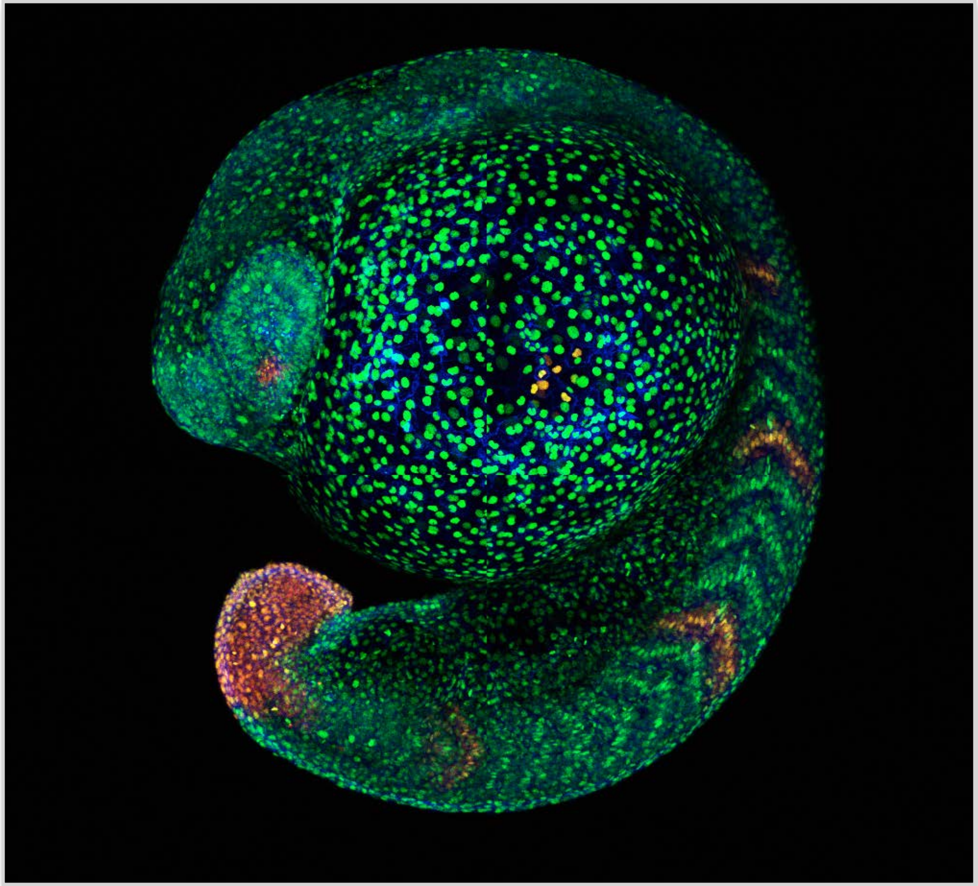
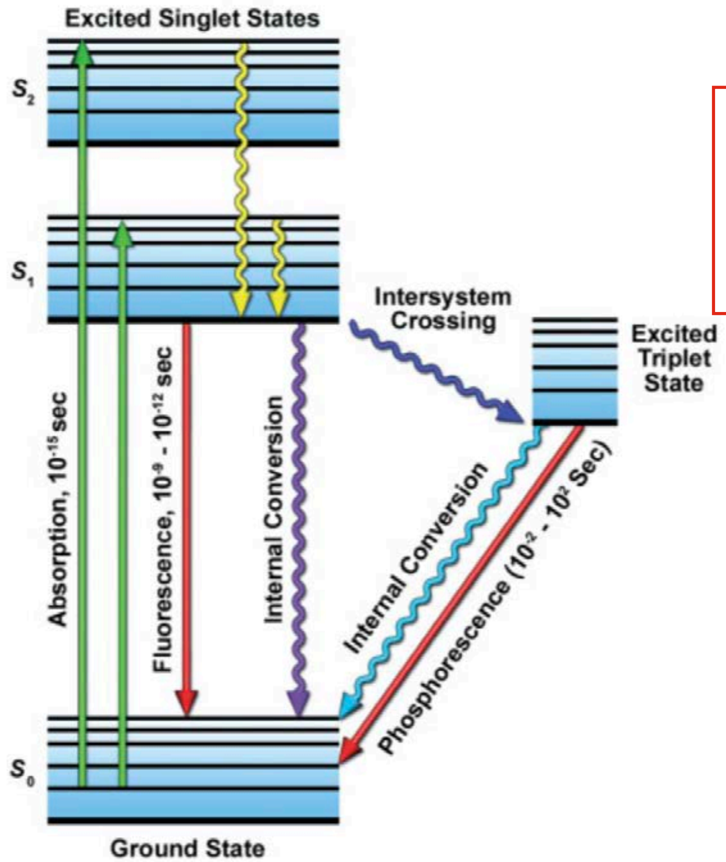
Fluorescence in biomedical research



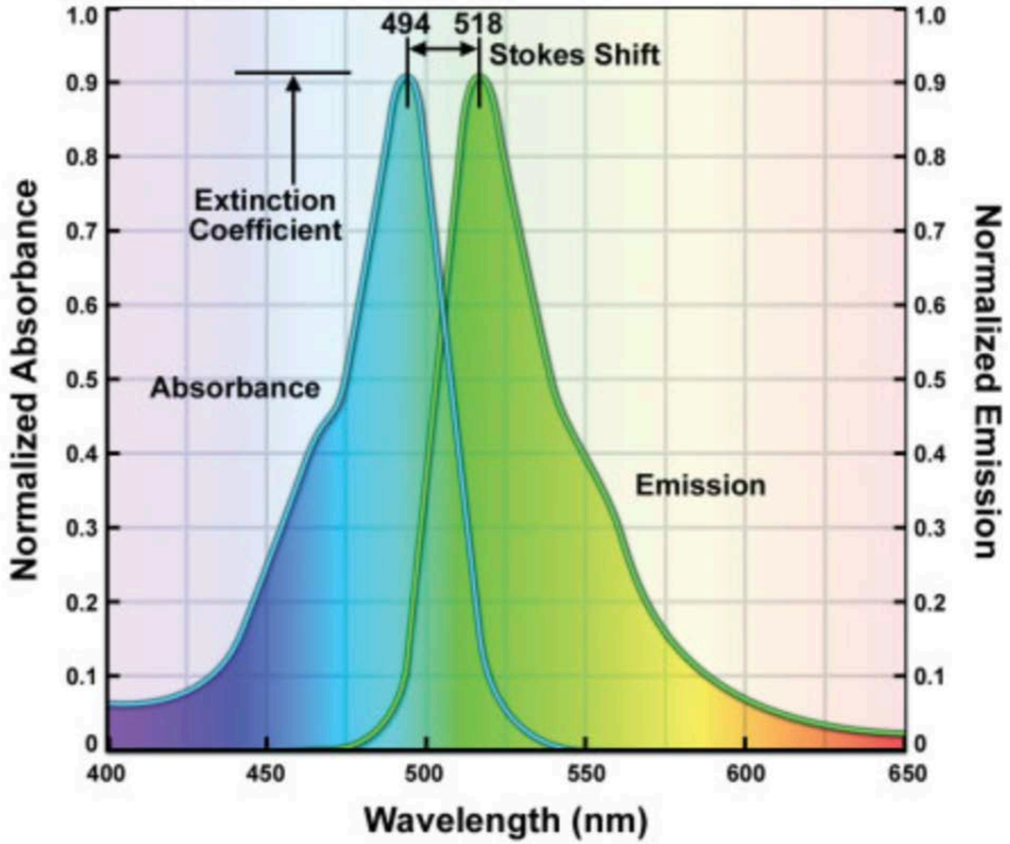
PLoS One 2012; 7(3):e32888.

Physical basis of fluorescence: The Perrin-Jablonski diagram



chemically reactive:
- photobleaching
- production of damaging radicals

Physical basis of fluorescence: The Stokes shift



Physical basis of fluorescence: Important Parameters

molar extinction coefficient
potential of a fluorochrome to absorb photon quanta (given in optical density) at a reference wavelength (usually the absorption maximum)

quantum efficiency (QE)
fraction of absorbed photon quanta that is re-emitted by a fluorochrome as fluorescent photons

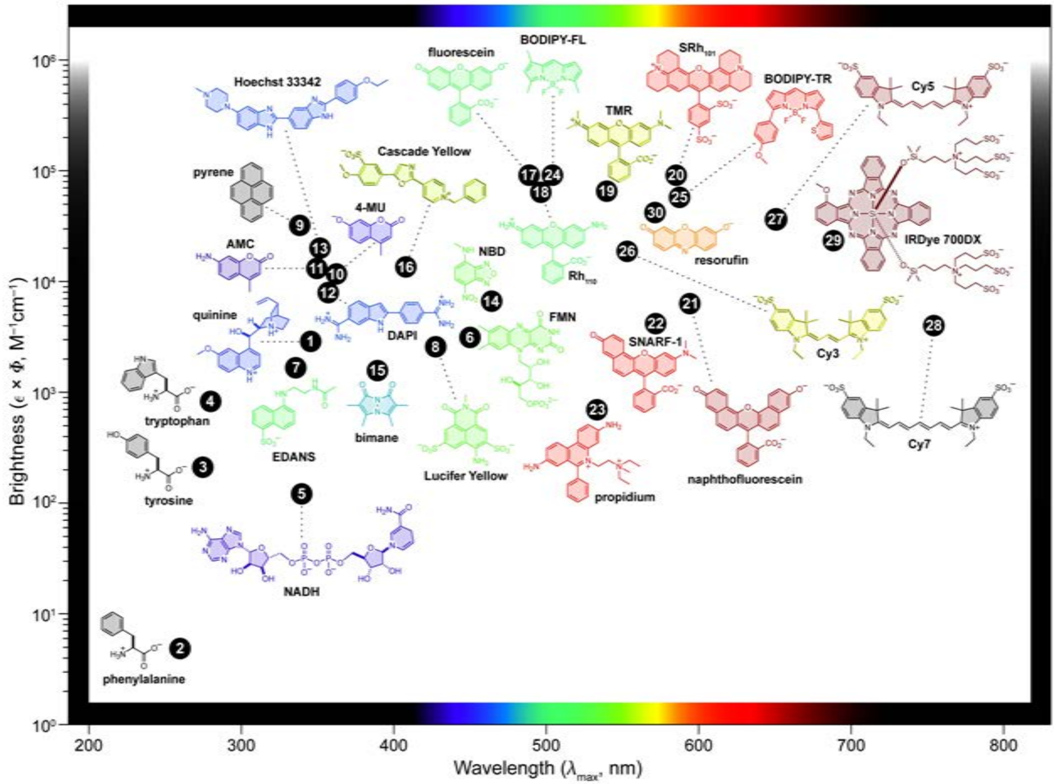
Quenching
reduction of quantum yield of a fluorochrome without changing its fluorescence emission spectrum; caused by interactions with other molecules including other fluorochromes

Photobleaching
permanent loss of fluorescence by a dye due to photon-induced chemical damage and covalent modification

Blinking
phenomenon that occurs during continuous excitation of a fluorescent molecule where emission transitions between “on” and “off” states

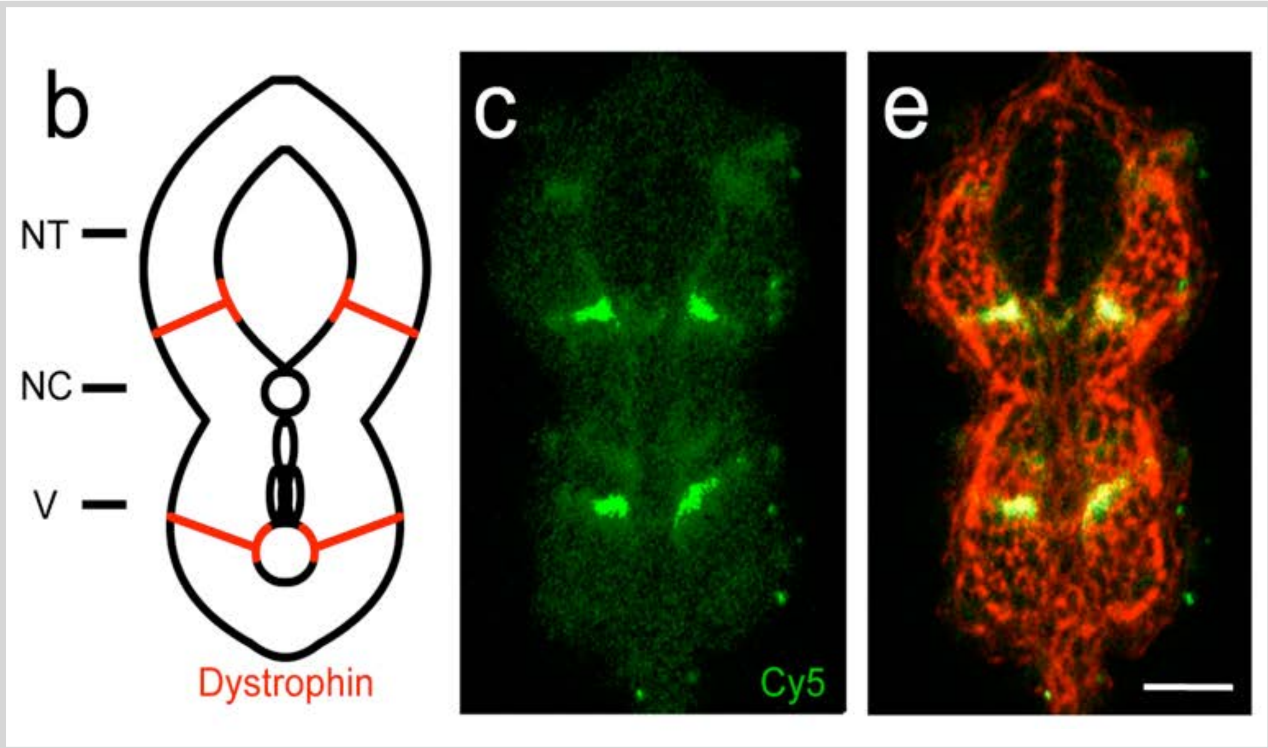
Properties of fluorescent dyes

BRIGHT - PHOTOSTABLE - COLOUR SELECTIVE - pH INSENSITIVE - WATER SOLUBLE



From Lavis, L.D., and Raines, R.T. 2008. ACS Chemical Biology 3:142–155

Autofluorescence of endogenous molecules



Proc Natl Acad Sci U S A. (2010); 107(33):14535-40.

What we will cover

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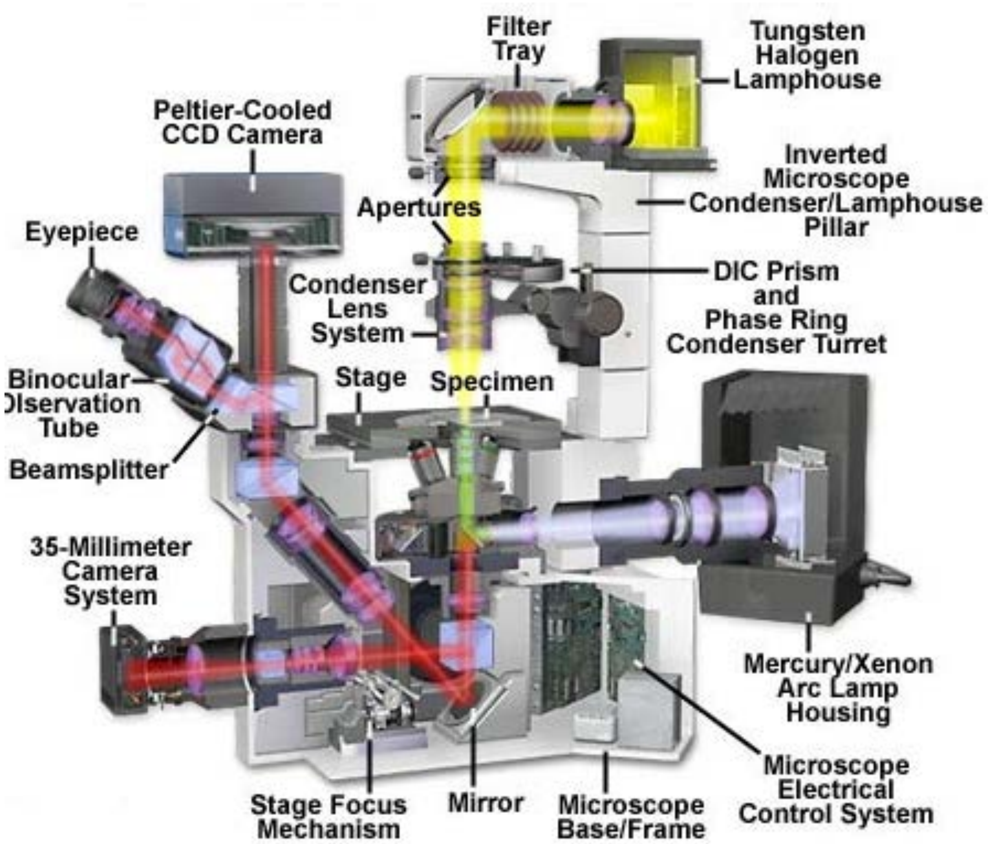
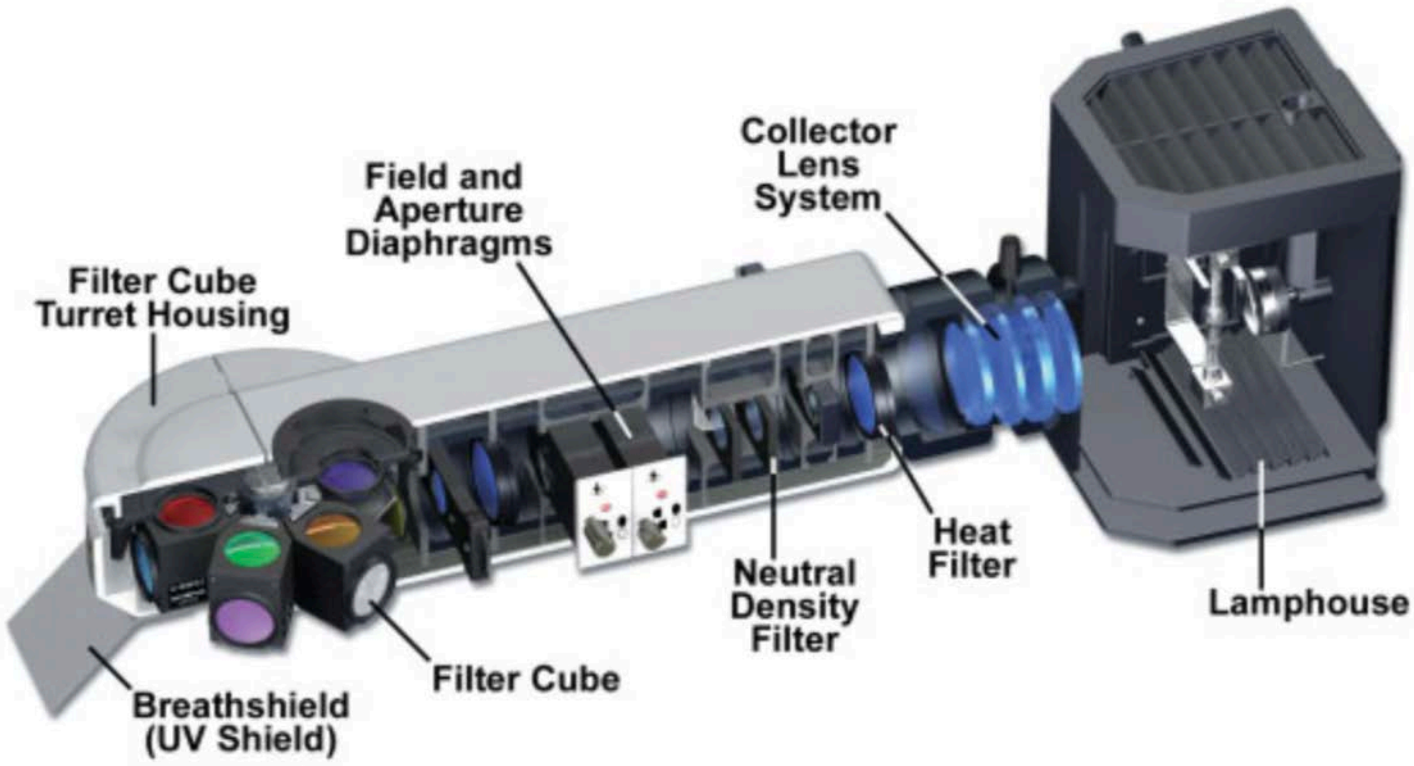
- What are genetically encoded fluorescent proteins?

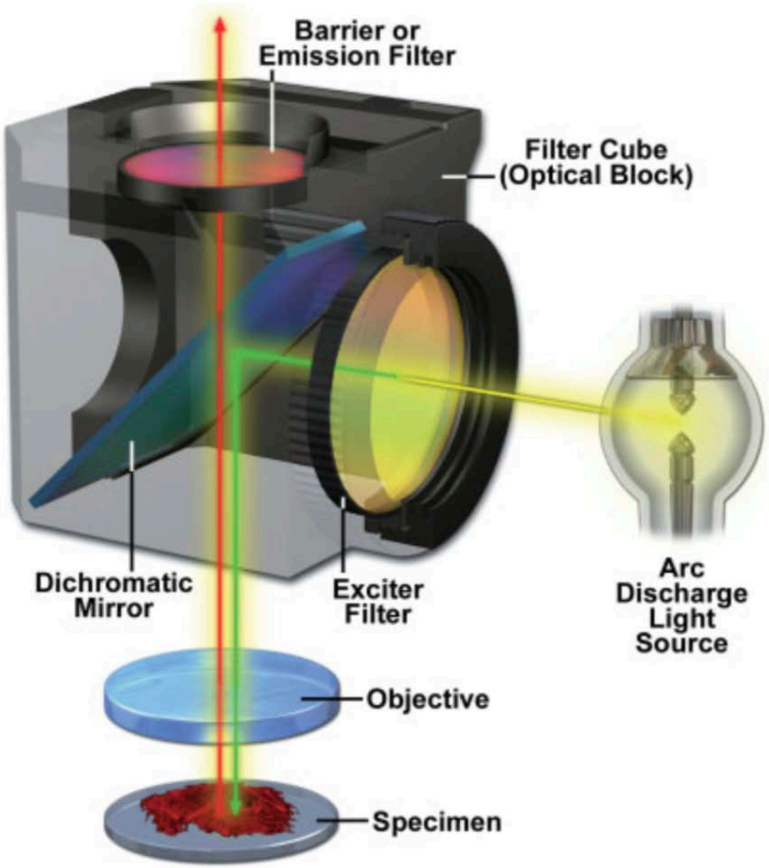
Arrangement of a fluorescent microscope



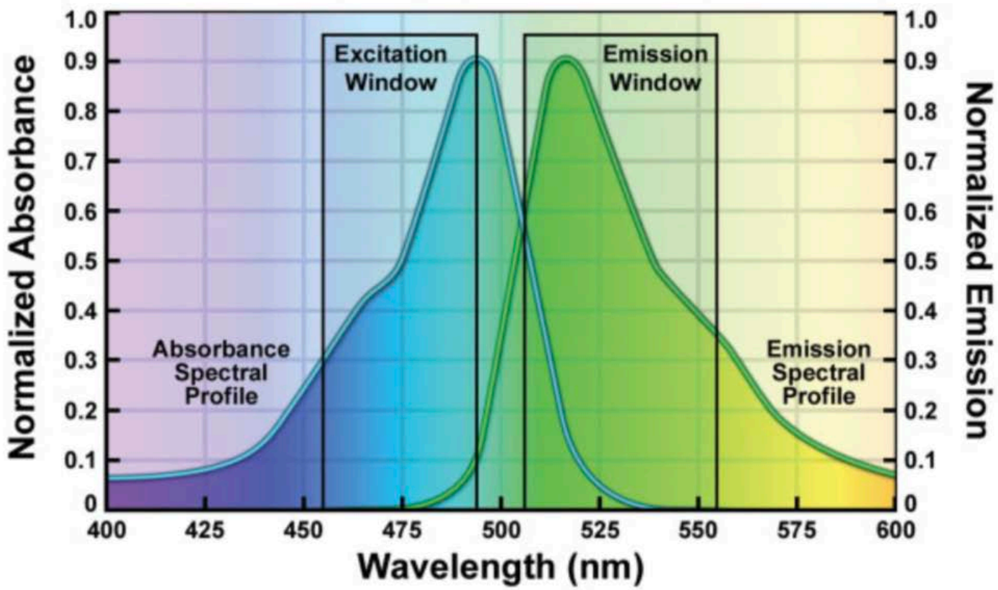
The filter cube turret housing in a fluorescent microscope

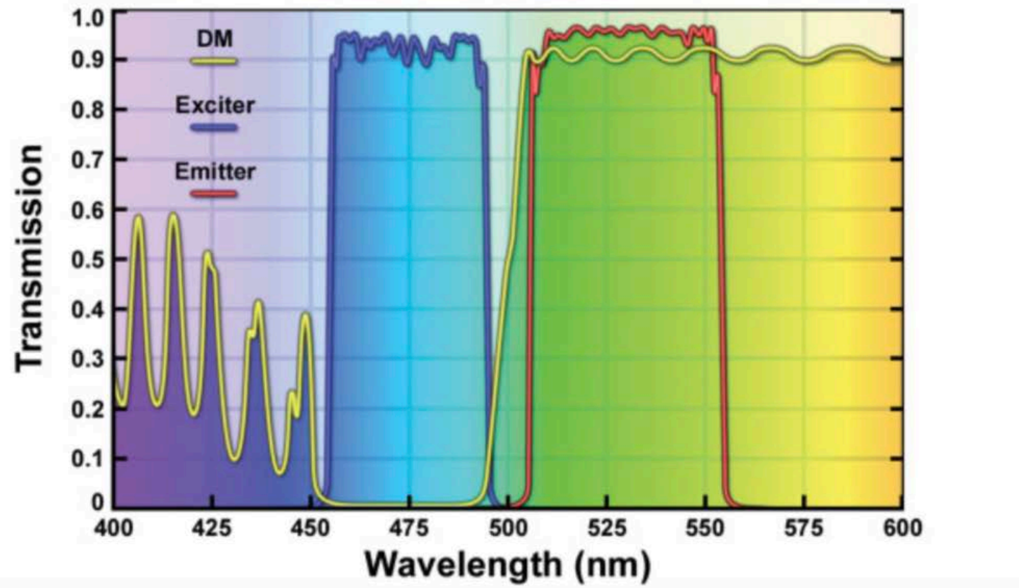


Arrangement of filters in a filter cube

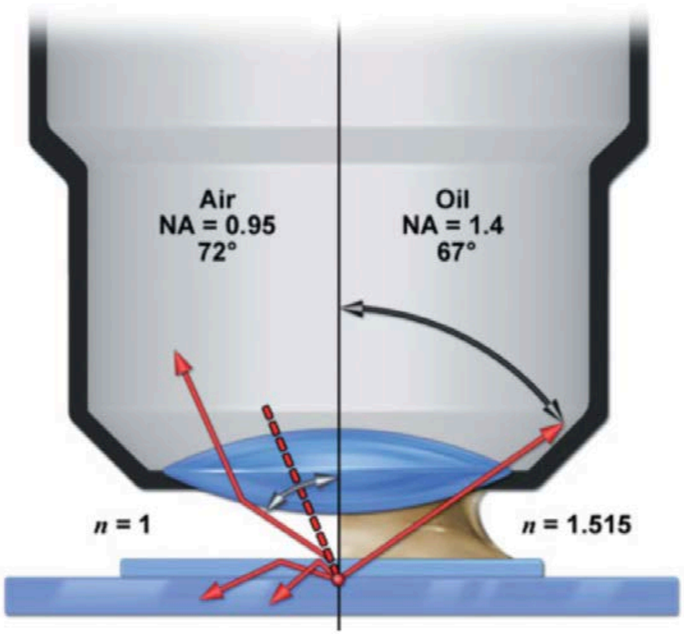


Transmission profiles of filters in a fluorescein filter set





Spatial resolution in fluorescent microscopy



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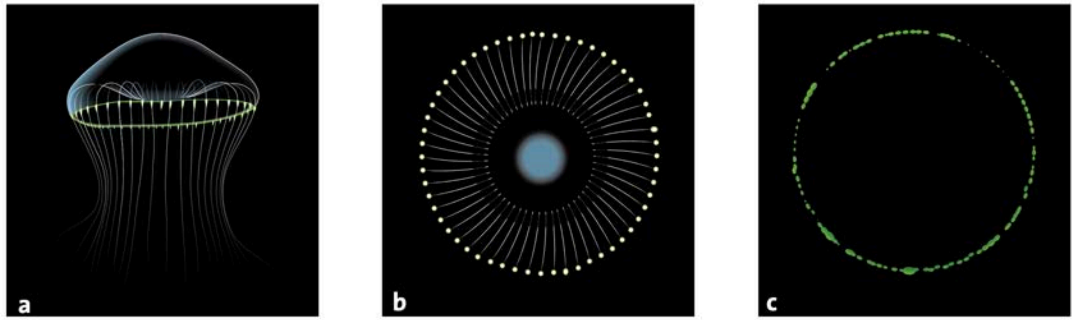
- **What are genetically encoded fluorescent proteins?**

Properties of fluorescent proteins

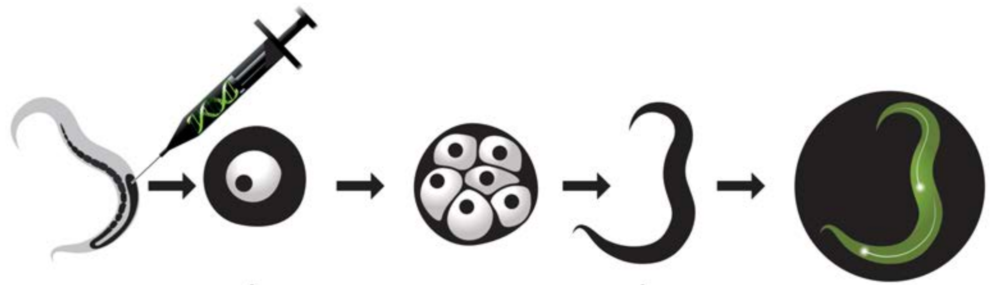


- 1960s and 1970s

  wt-GFP purified along with luminescent aequorin from Jellyfish *Aequorea Victoria*

- 1992 and 1994

  wt-GFP cloned and expressed

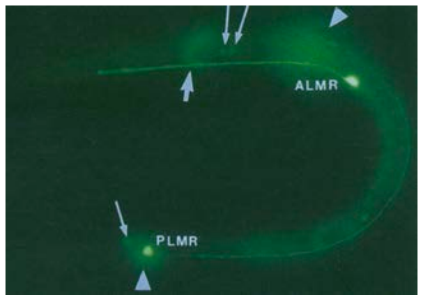


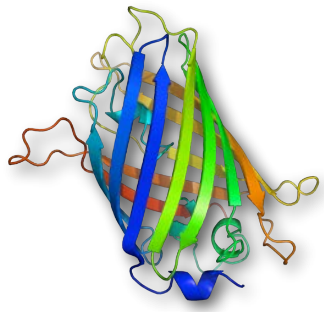
- 1995

  e-GFP introduced

- 1996

  wt-GFP/e-GFP crystal structure reported





Structure of a fluorescent proteins

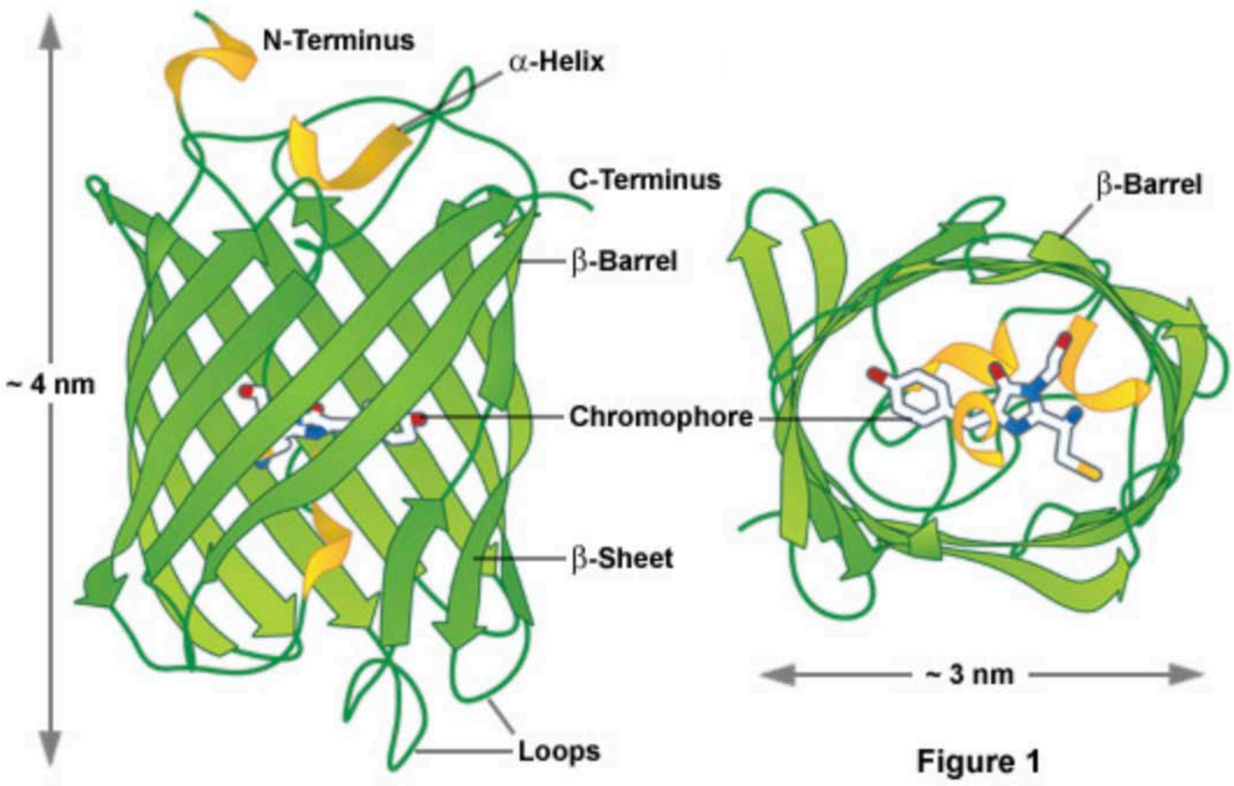
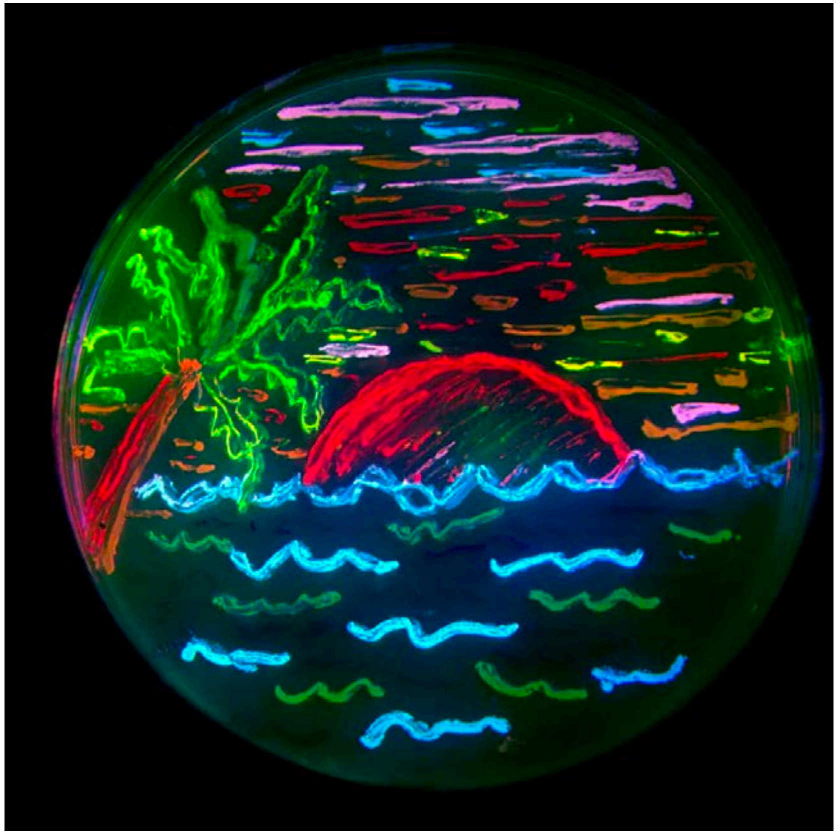


Figure 1

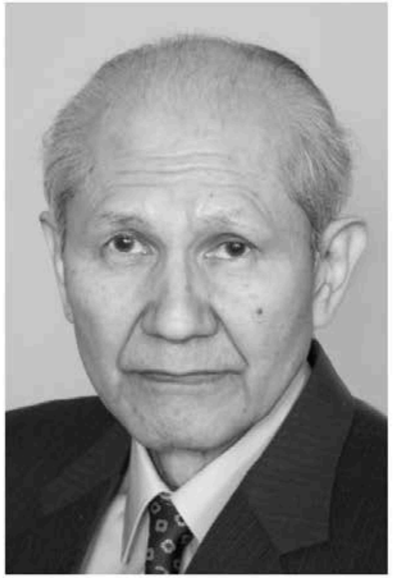
Differently coloured fluorescent proteins



Protein[a]	Color[b]	Excitation (nm)	Emission (nm)	Brightness[c]	Photostability[d]	Filter Set[e]
EBFP2	Blue	383	448	18	++	DAPI
mCerulean	Cyan	433	475	17	++	CFP
mTurquoise	Cyan	433	474	25	+++	CFP
mTFP1	Teal	462	492	54	+++	CFP
mEGFP	Green	488	507	34	++++	FITC/GFP
mEmerald	Green	487	509	39	++++	FITC/GFP
mVenus	Yellow	515	528	53	++	FITC/YFP
mCitrine	Yellow	516	529	59	++	FITC/YFP
mKO2	Orange	551	565	40	+++	TRITC
tdTomato	Orange	554	581	95	+++	TRITC
TagRFP	Orange	555	584	48	++	TRITC
mApple	Orange	568	592	37	+++	TRITC
mCherry	Red	587	610	17	+++	TxRed
mKate2	Far-Red	588	633	25	++	TxRed
mPlum	Far-Red	590	649	3.2	+++	TxRed
mNeptune	Far-Red	600	650	13	++++	Cy5

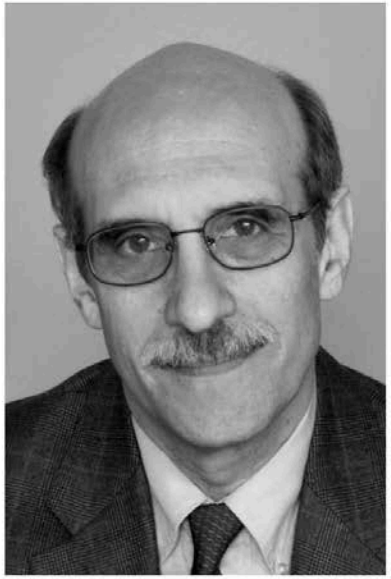
[a] Common literature abbreviation.
[b] Spectral class.
[c] Product of the molar extinction coefficient and the quantum yield $(mM \times cm)^{-1}$.
[d] Relative to mEGFP (++++).
[e] Recommended filter set.

The Nobel Prize in Chemistry 2008 was awarded "for the discovery and development of the green fluorescent protein, GFP."



Osamu Shimomura

Prize share: 1/3



Martin Chalfie

Prize share: 1/3

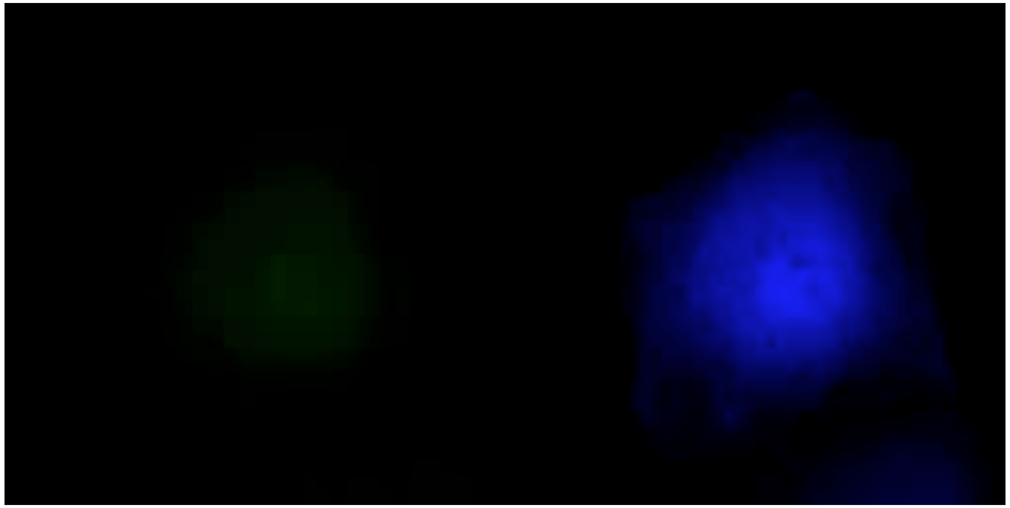
Roger Y. Tsien

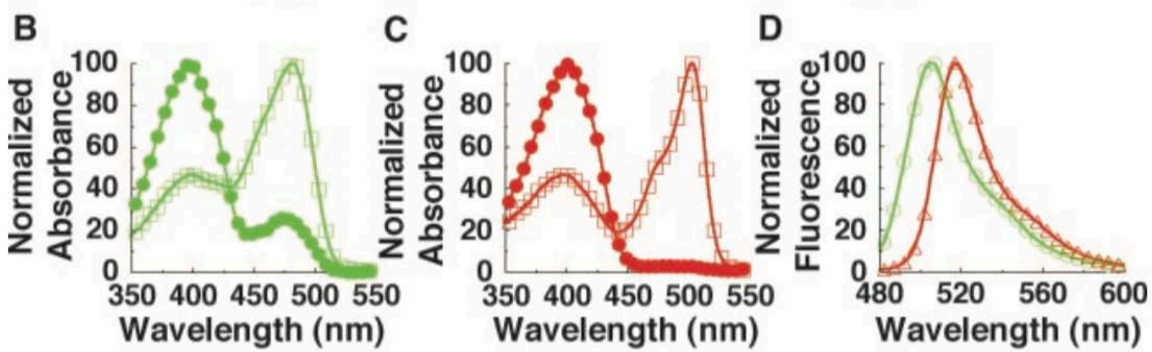
Prize share: 1/3

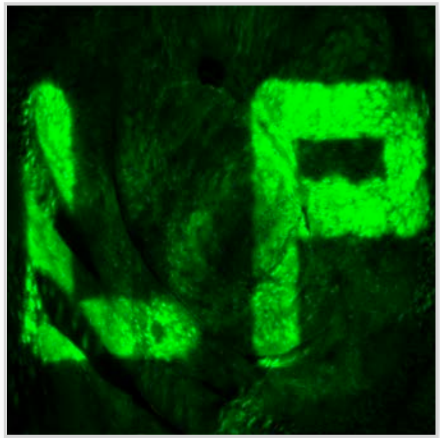
Comparison of fluorescent proteins versus fluorescent dyes

	Advantages	Disadvantages
organic dyes		
genetically encoded fluorophores		

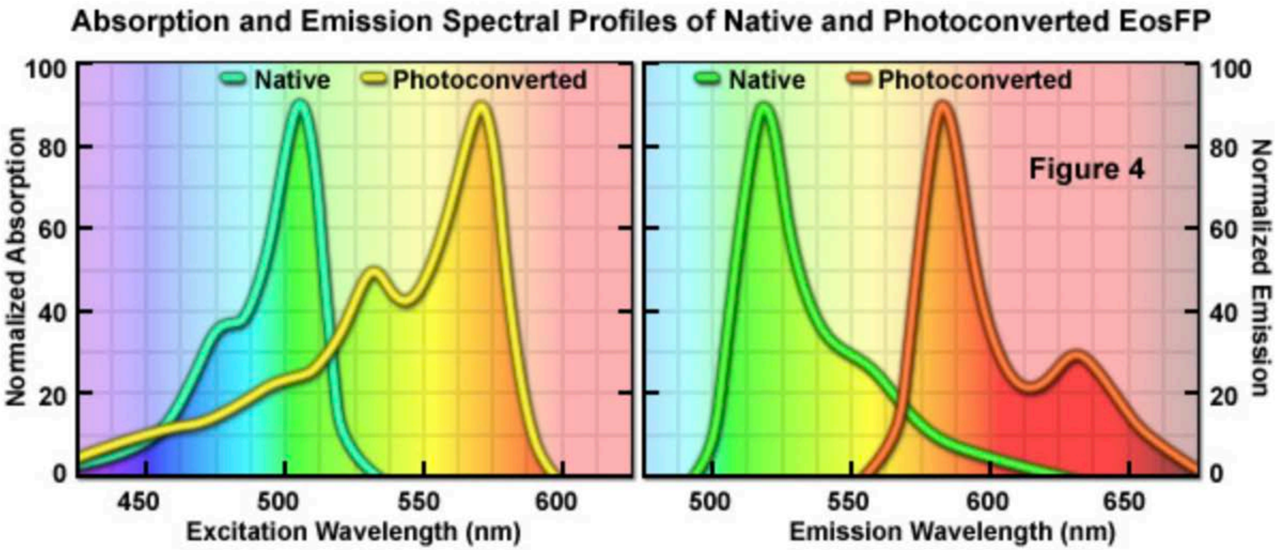
Photoactivatable fluorescent proteins

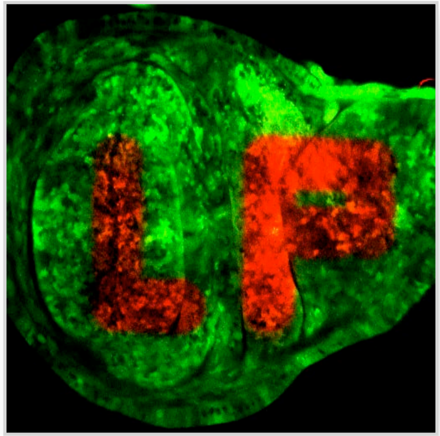




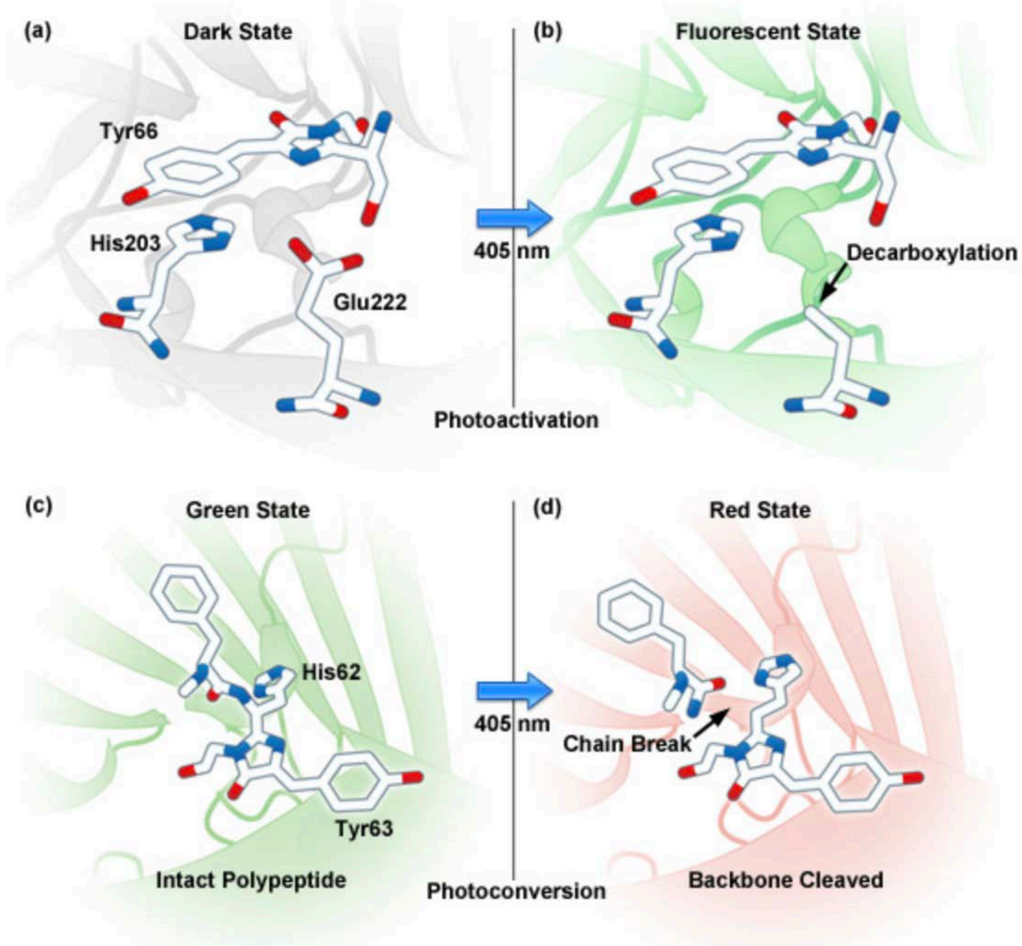


Photoconvertible fluorescent proteins





Mechanisms of photoactivation/photoconversion



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Thank you!

Periklis (Laki) Pantazis